

Donggeon (David) Oh

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Department of Electrical and Computer Engineering

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Website / Google Scholar / LinkedIn

RESEARCH INTERESTS

I push the boundaries of **safety assurance** for intelligent systems. In doing so,

- I integrate insights from **control theory**, **reinforcement learning (RL)**, and **dynamic game theory** to certify and enforce safety of high-dimensional, black-box systems under uncertainty.
- I investigate the **interplay between humans and robots** in safety-critical scenarios and how to foster safe, smooth, and **trustworthy collaboration**.

Ultimately, I envision a world where robots operate safely amidst other agents, including humans, in **unstructured, open-world environments**.

EDUCATION

Princeton University

2024–present

Ph.D. candidate in Electrical and Computer Engineering; expected May 2029.

Advisor: **Jaime Fernández Fisac**.

Seoul National University

2017–2024

B.S. in Aerospace Engineering & B.S. in Artificial Intelligence. Ranked 2nd of 71 students.

PUBLICATIONS

* Equal contribution. † Equal advising. ‡ Equal senior authorship.

Preprints

- [P2] Winning a won game: Strict reach–avoid–stay control barrier functions for high-dimensional black-box systems
D. D. Oh, D. P. Nguyen, G. Yuan, Q. Li, J. F. Fisac[†], H. Hu[†]
arXiv:2609.19449, 2026
- [P1] Turning safety into competence: Minimally exploitable robot policies via safety-filtered reinforcement learning
R. Wu*, R. Yang*, **D. D. Oh**, D. P. Nguyen[†], H. Hu[†]
arXiv:2609.27312, 2026

Peer-Reviewed Conference Publications

- [C7] Synthesis and deployment of maximal robust control barrier functions through adversarial reinforcement learning
D. D. Oh, D. P. Nguyen, H. Hu, J. F. Fisac
IEEE Conference on Decision and Control (CDC), 2026 (*to appear*)
- [C6] Unified overview of safety certificate functions: Techniques from Hamilton–Jacobi reachability and control barrier functions
H. Wang, M. Cohen, Z. Feng, **D. D. Oh***, O. So*, S. Zhang*, J. Borquez, C. Fan[†], J. F. Fisac[†], H. Hu[†], J. J. Choi[‡], S. Bansal[‡]
IEEE Conference on Decision and Control (CDC), 2026 (*Tutorial paper; to appear*)
- [C5] Provably optimal reinforcement learning under safety filtering
D. D. Oh*, D. P. Nguyen*, H. Hu, J. F. Fisac
Proceedings of IASEAI Conference, 2026
- [C4] Safety with agency: Human-centered safety filter with application to AI-assisted motorsports
D. D. Oh*, J. Lidard*, H. Hu, H. Sinhmar, E. Lazarski, D. Gopinath, E. S. Sumner, J. A. DeCastro, G. Rosman, N. E. Leonard, J. F. Fisac
Robotics: Science and Systems (RSS), 2025
- [C3] Safety-critical control under multiple state and input constraints and application to fixed-wing UAV
D. D. Oh*, D. Lee*, H. J. Kim
IEEE Conference on Decision and Control (CDC), 2023

- [C2] Stable contact guaranteeing motion/force control for an aerial manipulator on an arbitrarily tilted surface
J. Byun, B. Kim, C. Kim, **D. D. Oh**, H. J. Kim
IEEE International Conference on Robotics and Automation (ICRA), 2023
- [C1] Real-time trajectory generation of a quadrotor UAV with load suspended from a pulley
D. D. Oh, J. Byun, D. Lee
International Conference on Control, Automation and Systems (ICCAS), 2022

PROFESSIONAL ACTIVITIES

Workshop & Seminar Organization

- [W1] Workshop on safety analysis for autonomous system: Practical perspectives and recent advances
H. Wang, S. Zhang, O. So, J. F. Fisac, C. Fan, S. Bansal, Z. Feng, **D. D. Oh**, J. J. Choi, M. Cohen, H. Hu, J. Borquez
IEEE Conference on Decision and Control (CDC), 2026 (*upcoming*)

Reviewer

IEEE Robotics and Automation Letters (RA-L); IEEE Conference on Decision and Control (CDC); International Association for Safe & Ethical AI (IASEAI); International Conference on Neuro-symbolic Systems (NeuS); ICRA Workshop on Public Trust in Autonomous Systems (PTAS).

INVITED TALKS

- [T1] Towards safety for black-box systems: Application to AI-assisted racing and safe RL
Safe and Intelligent Autonomy (SIA) Lab, Stanford University, Sept. 8, 2026.

PATENT APPLICATIONS

- [PA1] Safety system for a vehicle
N. Leonard, J. DeCastro, G. Rosman, J. F. Fisac, **D. D. Oh**, J. Lidard, H. Hu, D. Nguyen
U.S. Patent Application No. 19/641,805. Filed Apr. 2026. *Patent pending*.

TEACHING & MENTORING

Head Teaching Assistant, Intelligent Robotic Systems Spring 2026
Princeton University. Instructor: Jaime Fernández Fisac.

Research mentorship

Elle Lazarski (mentored as a Princeton undergraduate; now a Ph.D. student at UC Berkeley).

HONORS & AWARDS

Presidential Science Scholarship, Republic of Korea 2017–2022
National scholarship recognizing 120 top STEM students.

Talent Award of Korea 2016
National honor recognizing 100 outstanding Koreans across disciplines.

RESEARCH METHODS & TOOLS

Computation: Python, PyTorch, MATLAB/Simulink, C/C++, Linux, Git; MuJoCo, Safety Gymnasium.

Human-in-the-loop research: Authored IRB protocol and study materials and conducted an 83-participant driving study in Assetto Corsa; wheel/pedal interface with a custom Linux force-feedback torque driver.

REFERENCES

Jaime Fernández Fisac, Princeton University jfisac@princeton.edu
Assistant Professor of Electrical and Computer Engineering.

Naomi Ehrich Leonard, Princeton University naomi@princeton.edu
Chair and Edwin S. Wilsey Professor of Mechanical and Aerospace Engineering.

Haimin Hu, Johns Hopkins University haimin@cs.jhu.edu
Assistant Professor of Computer Science.